

Mobile HCI

Lecture 2: Design principles and interaction techniques

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Agenda

- Recap on Mobile challenges for designing interaction
- Design Principles
- Interaction qualities and interaction techniques

Challenges (Benyon)

Chapter 19 (Mobile Computing) in **Designing interactive systems: A comprehensive guide to HCI, UX and interaction design** by Benyon, David. Pearson, 2014.



Because of the small screen it is difficult to achieve the design principle of visibility.



Functions have to be tucked away and accessed by multiple levels of menu, leading to difficulties of navigation.



Another feature is that there is not room for many buttons, so each button has to do a lot of work. This results in the need for different 'modes' and this makes it difficult to have clear control over the functions.



Visual feedback is often poor and people have to stare into the device to see what is happening.



Thus other modalities such as sound and touch are often used to supplement the visual.

Challenges of designing for mobile interactions

- **Reduced screen real estate:** less space for content and menus, leading to *discoverability* issues
- **Occlusion & Fat Finger Problem:** our fingers interact directly with the screen, *occluding* the content
- **Variable Contexts of Use:** users are more prompt to interruptions and situational impairments (e.g., can't type because it's freezing cold outside!)
- **Limited Input Expressiveness:** without the fine-grained control of a mouse or the combinatorial power of a physical keyboard, mobile devices have a narrower range of potential interactions (e.g., tap, double tap, swipe...), overloading the same interaction with many potential commands

- Zhanna Sarsenbayeva, Niels van Berkel, Chu Luo, Vassilis Kostakos, and Jorge Goncalves. 2017. Challenges of situational impairments during interaction with mobile devices. In Proceedings of the 29th Australian Conference on Computer-Human Interaction (OzCHI '17). Association for Computing Machinery, New York, NY, USA, 477–481. <https://doi.org/10.1145/3152771.3156161>
- Jessalyn Alvina, Carla F. Griggio, Xiaojun Bi, and Wendy E. Mackay. 2017. CommandBoard: Creating a General-Purpose Command Gesture Input Space for Soft Keyboard. In Proceedings of the 30th Annual ACM Symposium on User Interface Software and Technology (UIST '17). Association for Computing Machinery, New York, NY, USA, 17–28. <https://doi.org/10.1145/3126594.3126639>
- David Benyon (2019). Designing User Experience: A guide to HCI, UX and interaction design (4. ed), Pearson Education Limited.

Design Principles

Technology-independent strategies for
describing and *generating* interactions

Reification, polymorphism and reuse

Three design principles for *simple but powerful* interactions.

Reification:

turning an *abstract concept* or *action* into a *concrete, manipulable object* in the interface

Polymorphism:

allowing the *same command* to affect *different types* of objects

Reuse:

recycling *past* interactions to optimize *future* actions

Michel Beaudouin-Lafon and Wendy E. Mackay. Reification, polymorphism and reuse: three principles for designing visual interfaces. AVI '00. <https://doi.org/10.1145/345513.345267>

Wendy E. Mackay and Michel Beaudouin-Lafon. 2025. Interaction Substrates: Combining Power and Simplicity in Interactive Systems. In Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems (CHI '25). <https://doi.org/10.1145/3706598.3714006>

You can hear Michel talk about these principles himself here: https://www.youtube.com/watch?v=NvB_2vL1UmA

Reification

Turning an *abstract concept* or *action* into a concrete, manipulable *object* in the interface.

This means that you can *interact* with the object, for example, to *modify* it.

Example: the results of the action “highlighting” are turned into *highlight* objects that I can interact with

Highlights in Preview (Mac)

Highlights and Notes

Identifying the Values that Shape HCI and CSCW Research in Latin American Communities: A Collaborative Autoethnography

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ABSTRACT

Over the past decade, community collaborations have come into focus within the HCI and CSCW fields. Largely the result of increased concern for social and contextual dimensions of practice, these partnerships facilitate a pathway for researchers and practitioners to foreground the nuances of technology as it takes place in the real world. How these collaborations are engaged, what values mediate them, and how practices might vary across geographies remain active research questions. In this paper, we contribute by zooming into the experience of four HCI and CSCW researchers engaging in community collaborations in Latin America (LATAM). Through a collaborative autoethnography (CAE), we identify three main value tensions impacting HCI practices and methods in research collaborations with LATAM communities: camaraderie vs. cautiousness, informality vs. formality and hopefulness vs. transparency. Building on our findings, we provide three recommendations for researchers interested in engaging in community-based research in similar contexts.

*These authors contributed equally to this research and share first authorship.

CCS CONCEPTS

- **Human-centered computing** → HCI theory and models.

KEYWORDS

HCI; Latin America; Social Justice; Autoethnography; Critical HCI

ACM Reference Format:

Carla F. Griggio, Mayra D. Barrera Machuca, Marisol Wong-Villalacr, Laura S. Gaytán-Lugo, Karla Badillo-Urquiola, Adriana Alvarado, Monica Perusquia-Hernandez, Marianela Ciolfi Felice, Franceli L. Cibrian, Michaelanne Thomas, Carolina Fuentes, and Pedro Reynolds-Cuellar. 2024. Identifying the Values that Shape HCI and CSCW Research with Latin American Communities: A Collaborative Autoethnography. In *Conference on Computer-Supported Cooperative Work and Social Computing (CSCW Companion '24)*, November 9–13, 2024, San Jose, Costa Rica. ACM, New York, USA, 8 pages. <https://doi.org/10.1145/3678884.3681904>

1 INTRODUCTION

Community collaborations have emerged as an important research focus within the Human-Computer Interaction (HCI) and Computer-Supported Collaborative Work (CSCW) fields over the past few decades, partly under the umbrella of what has been referred to as the “third wave of HCI” [12]. Largely a result of the role played by context against a backdrop of ubiquitous computing, community-collaborations have been shaped by the changing nature of designing and deploying technology in the real world. This has led to a growing interest in understanding the values that shape these collaborations and how they might be supported or hindered by different contexts and technologies.

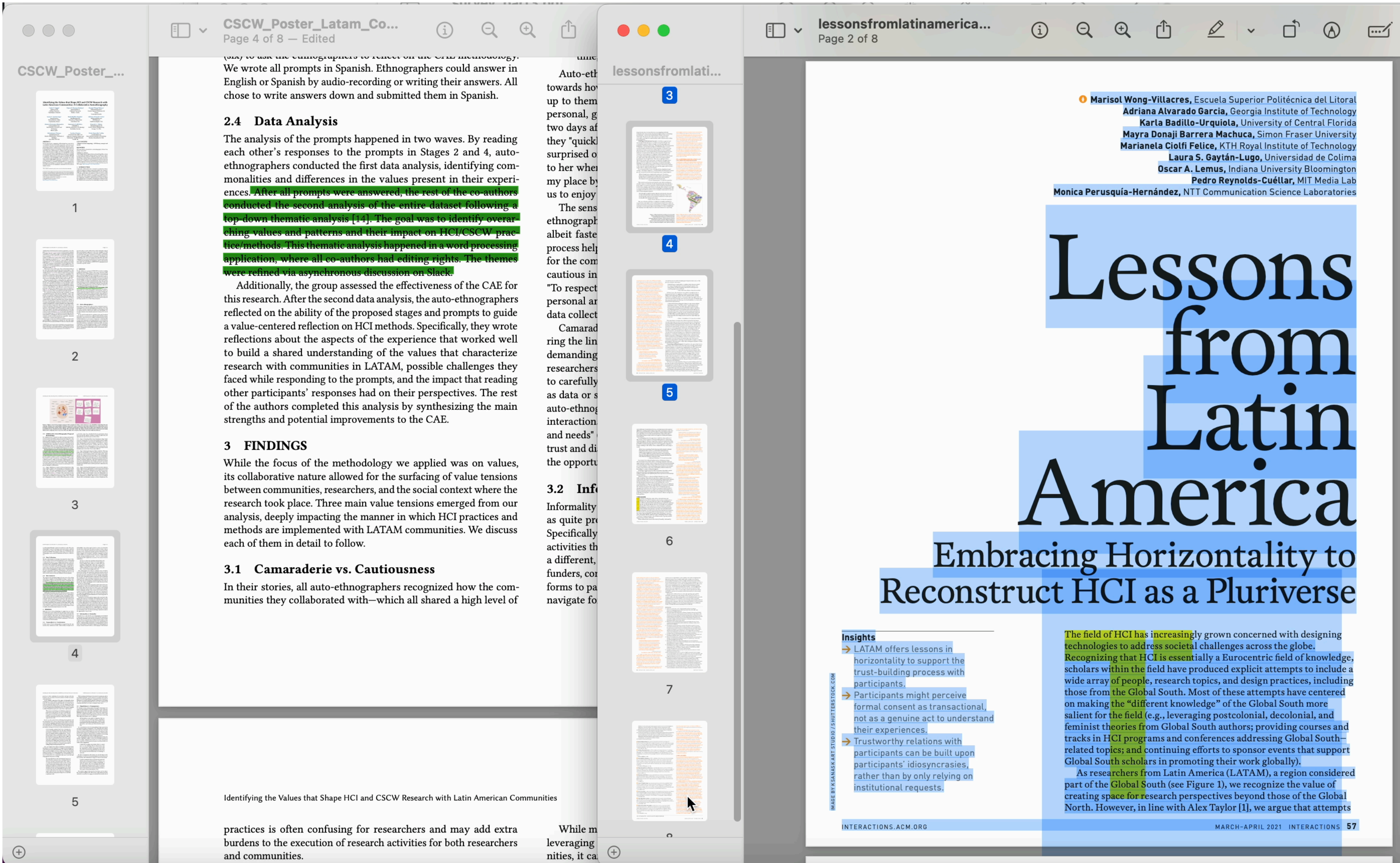
Reification

Turning an *abstract concept* or *action* into a concrete, manipulable *object* in the interface.

This means that you can *interact* with the object, for example, to *modify* it.

Example: the abstract concept of a “page” turns into page thumbnails that I can reorder and drag-and-drop across documents

Pages in Preview (Mac)

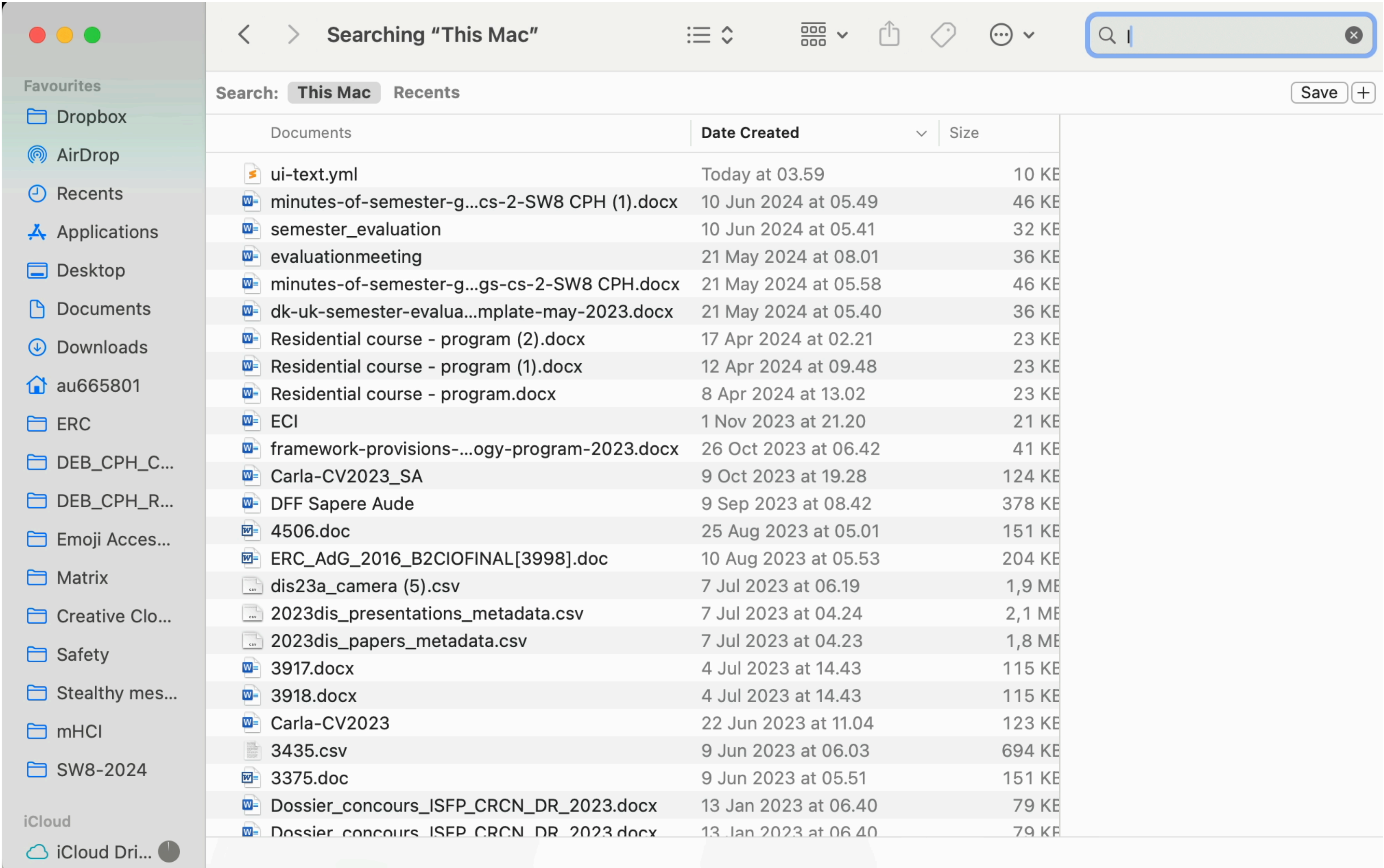


Reification

Allowing a *command* to affect *different* Turning an *abstract concept* or *action* into a concrete, manipulable *object* in the interface.

This means that you can *interact* with the object, for example, to *modify* it. of objects.

Example: the action of “searching and filtering” turns into a smart folder object



Finder en Mac

Let's go *mobile*

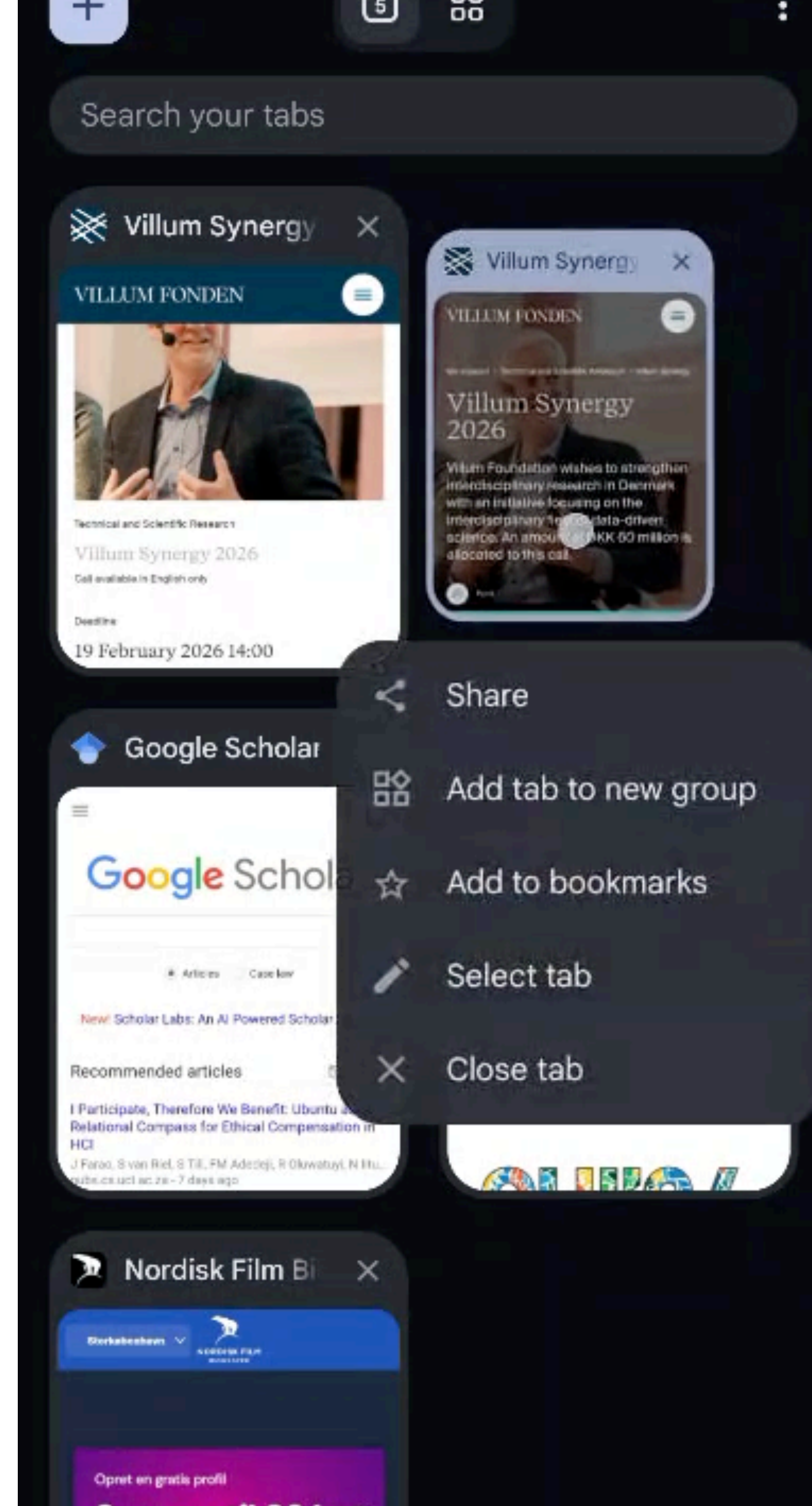
Reification

Turning an *abstract concept* or *action* into a concrete, manipulable *object* in the interface.

This means that you can *interact* with the object, for example, to *modify* it.

Example: turning the action (or intention) of "Come back here later" when navigating the web into a tab object

Tabs can be accessed later, reordered, closed, shared, etc.



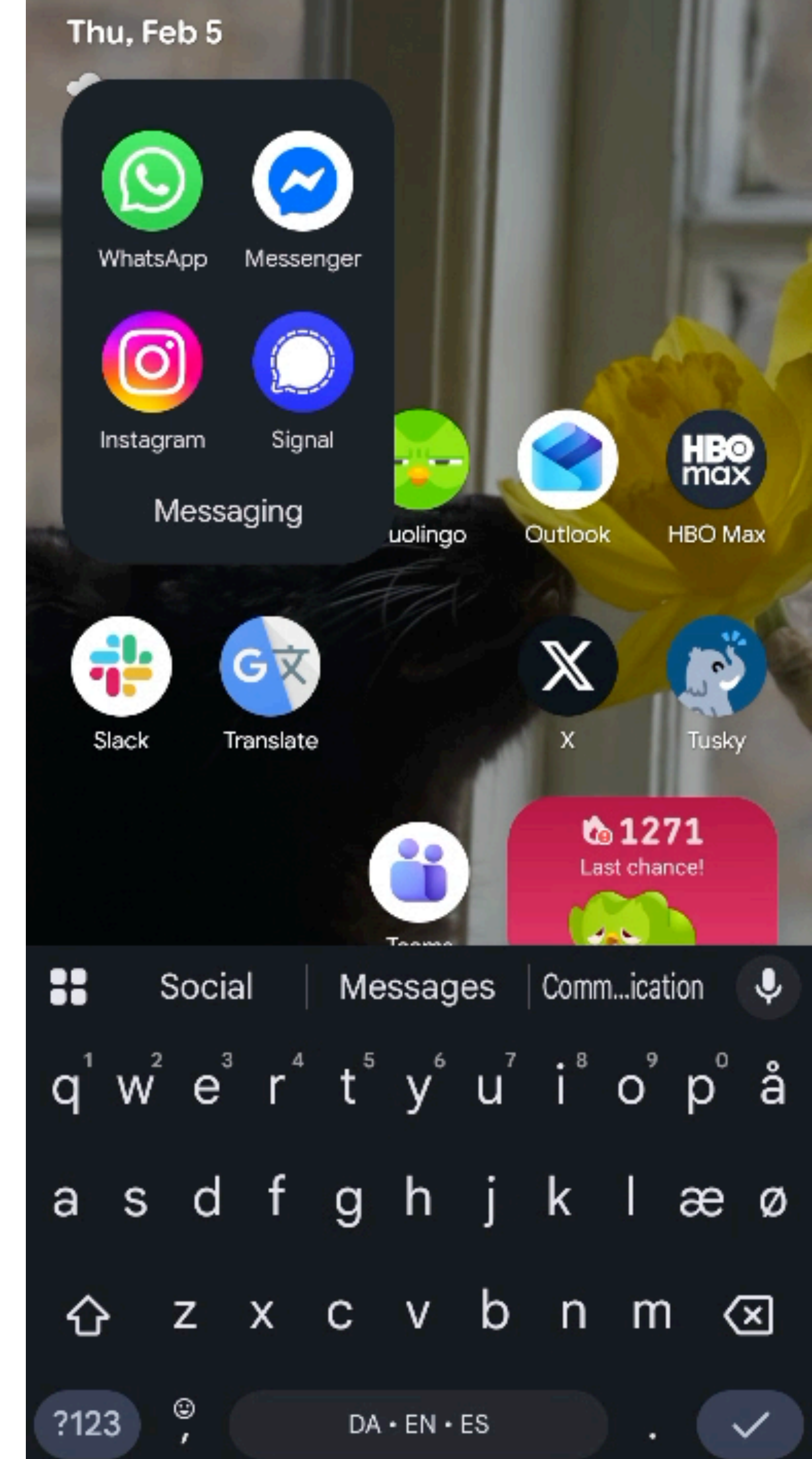
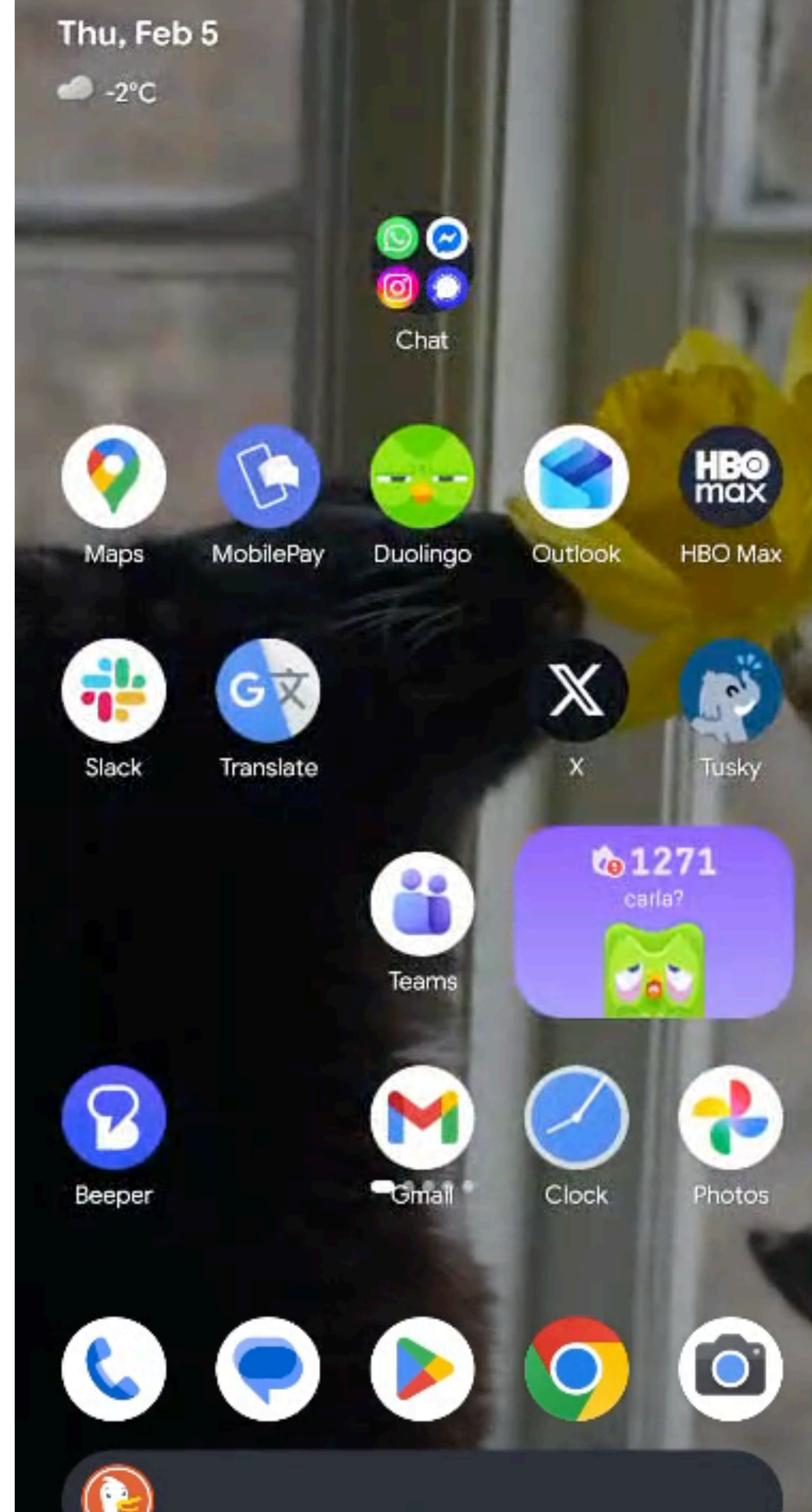
Reification

Turning an *abstract concept* or *action* into a concrete, manipulable *object* in the interface.

This means that you can *interact* with the object, for example, to *modify* it.

Example: turning the abstract concept of “a group of related apps” into a concrete object (an app group) that can be named.

App Groups (Android)

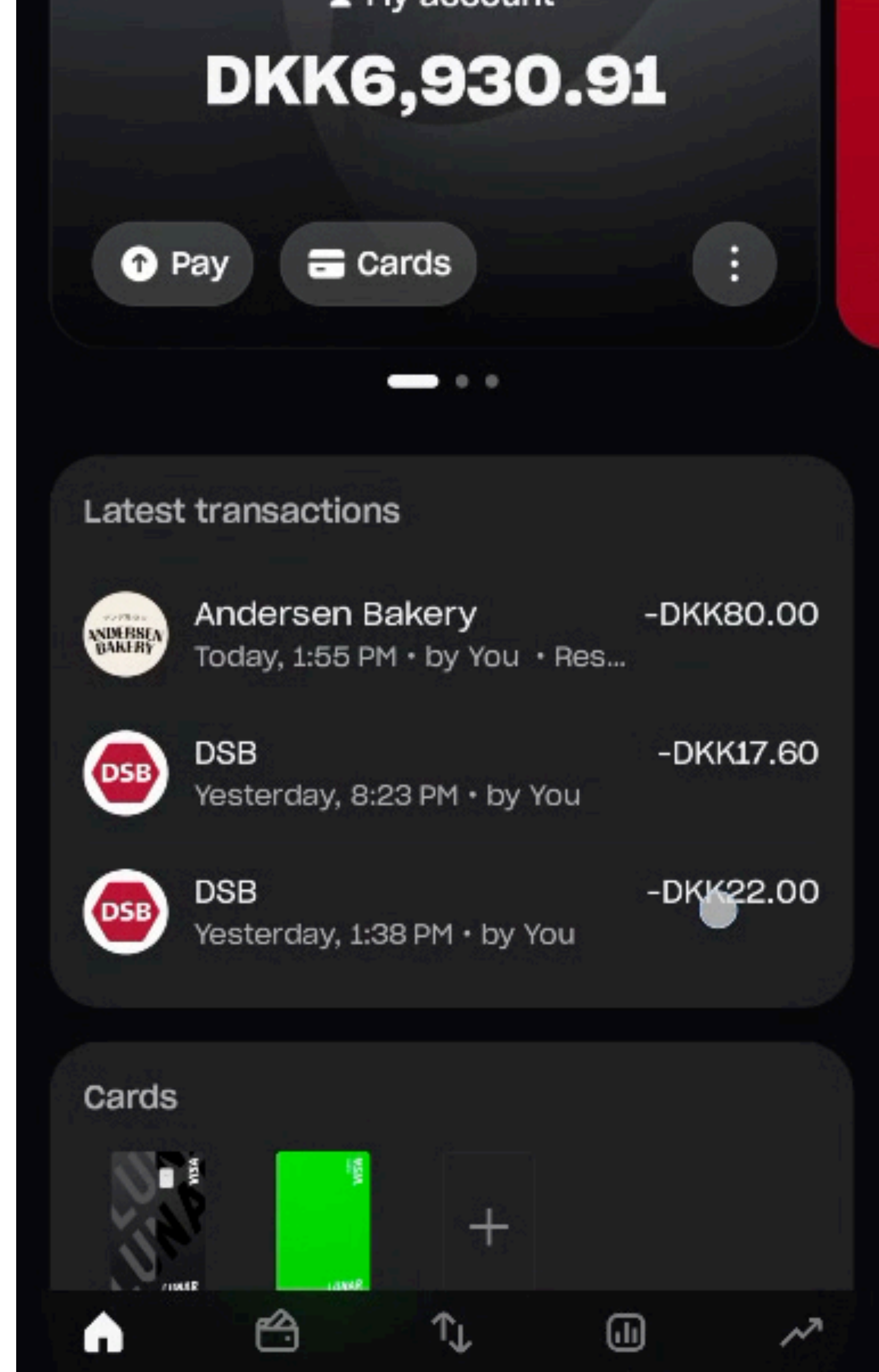


Reification

Turning an *abstract concept* or *action* into a concrete, manipulable *object* in the interface.

This means that you can *interact* with the object, for example, to *modify* it.

Example: instead of simply listing the information of every transaction, the transaction becomes an object with modifiable properties.



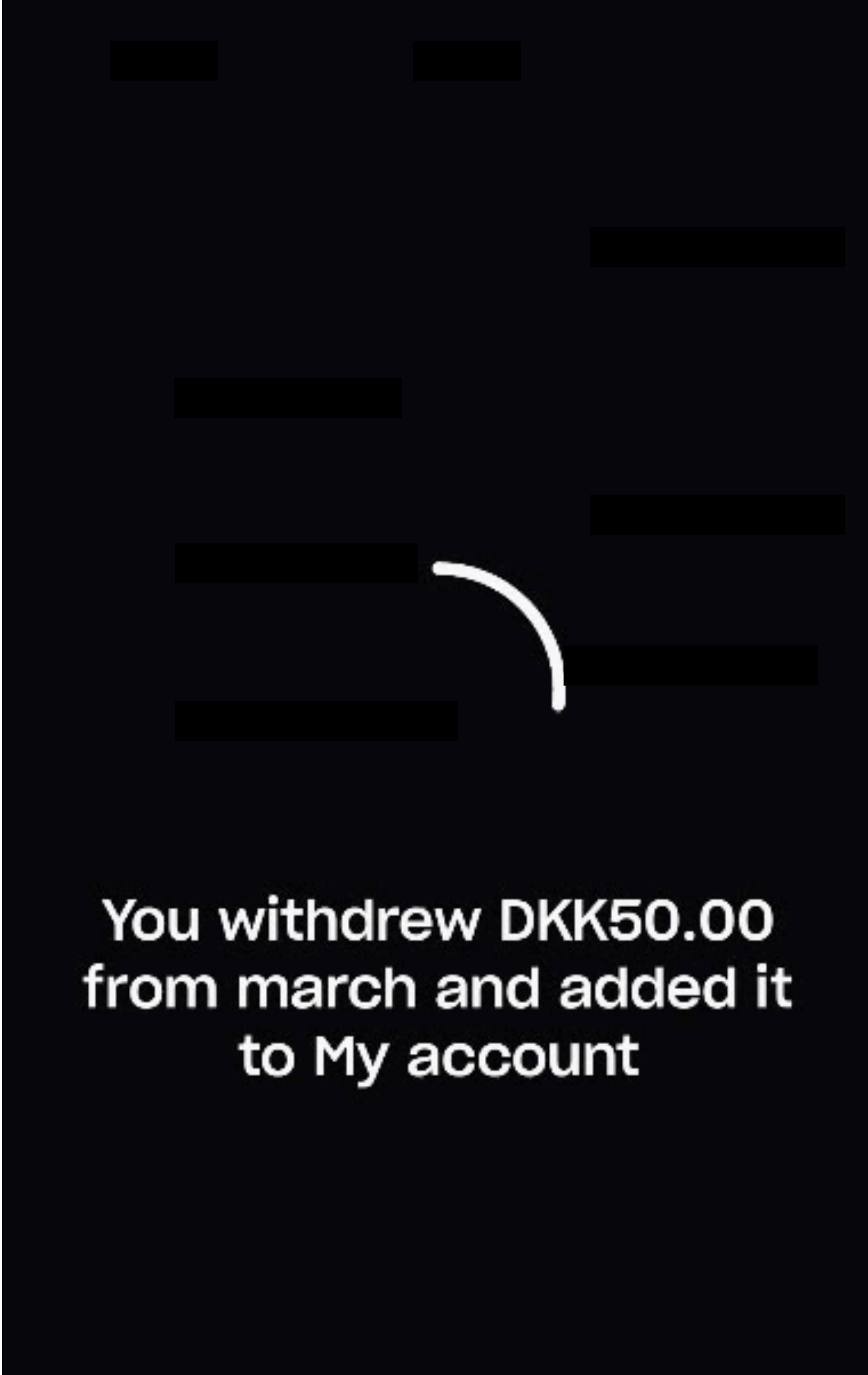
Reification

Turning an *abstract concept* or *action* into a concrete, manipulable *object* in the interface.

This means that you can *interact* with the object, for example, to *modify* it.

The reified action can become the input to other commands... *or the command itself!*

Dragging a “savings goal” to an account triggers the “withdraw money” command



You withdrew DKK50.00
from march and added it
to My account

Reification

Other state-of-the-art examples outside of mobile devices



Zoom

What if instead of having fixed “talking heads” on video calls...
we could manipulate each video feed?



Grønbæk et al. MirrorBlender: Supporting Hybrid Meetings with a Malleable Video-Conferencing System. CHI '21. <https://doi.org/10.1145/3411764.3445698>

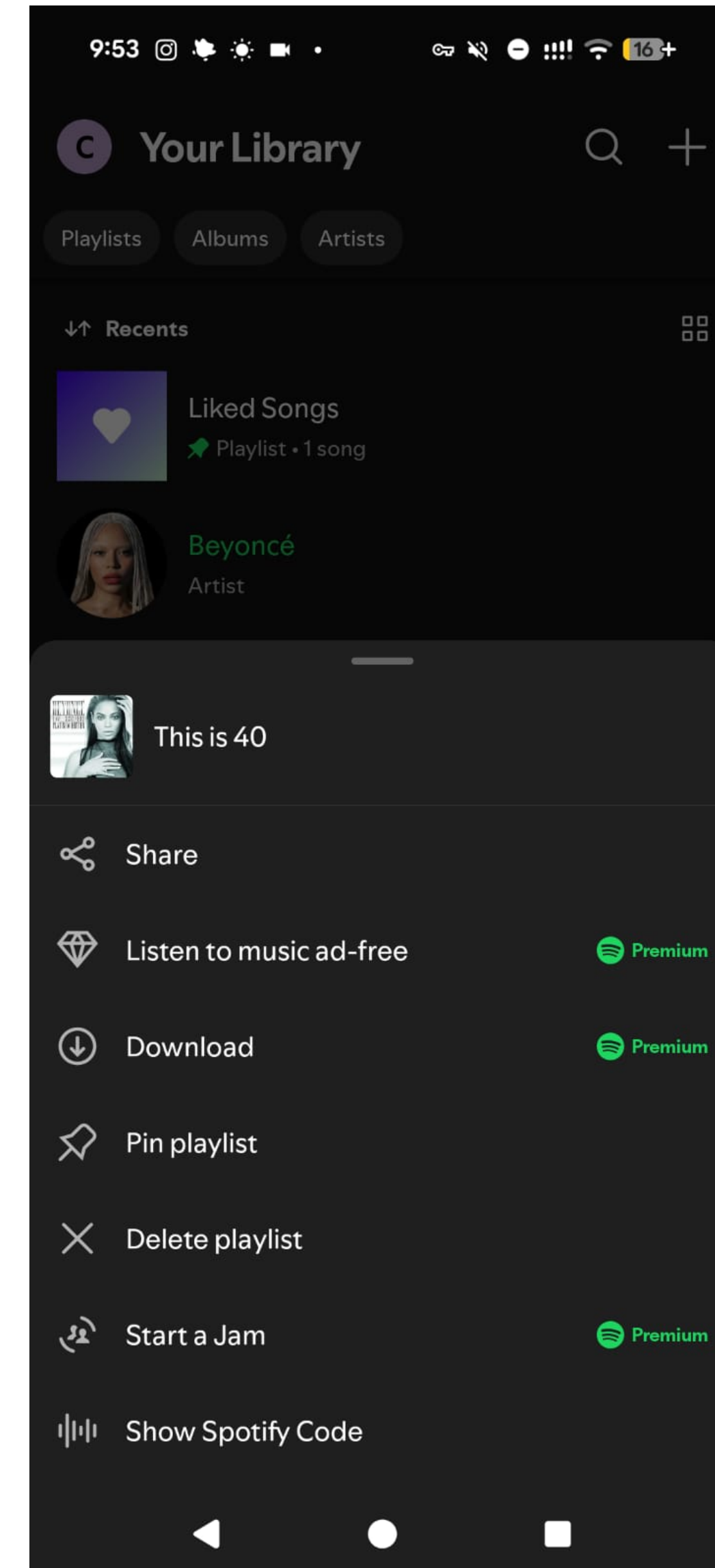
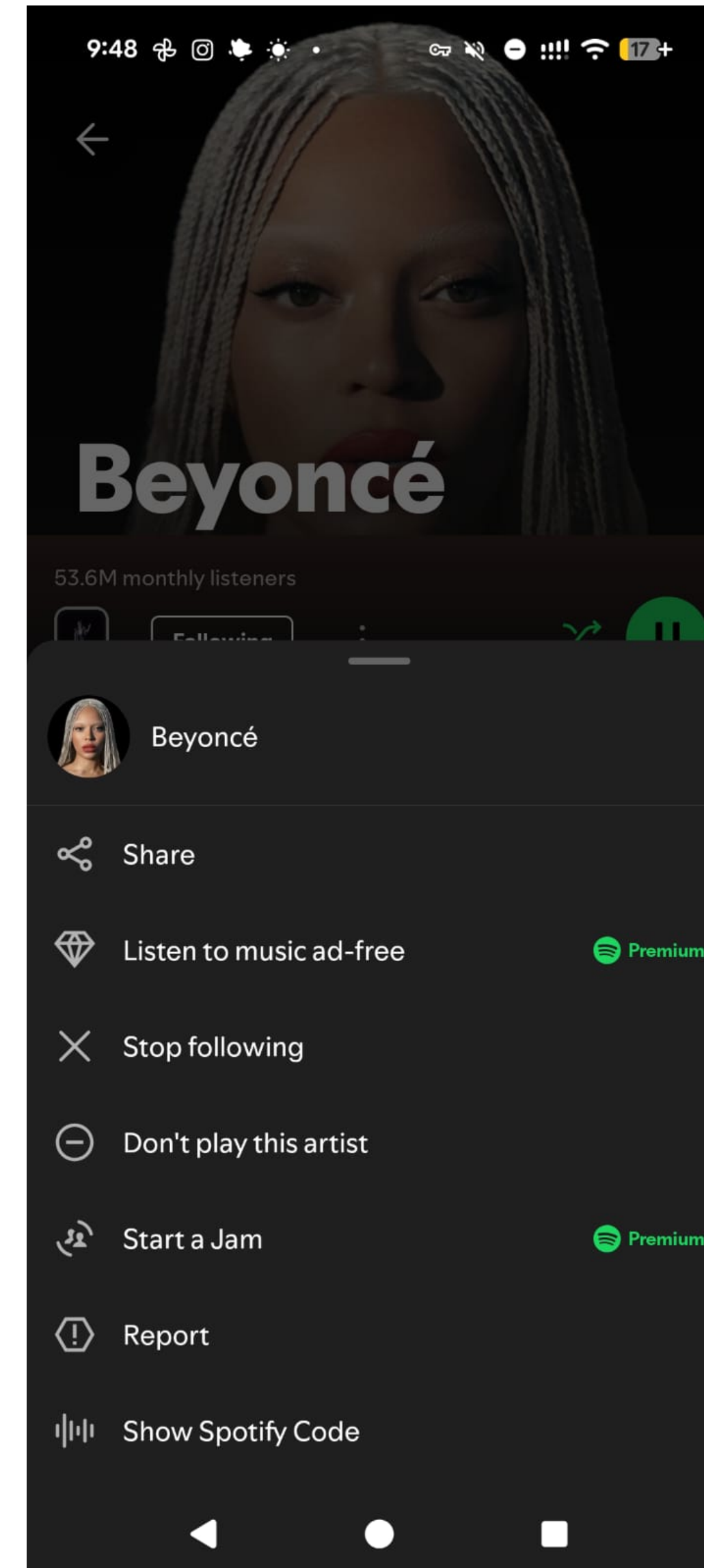
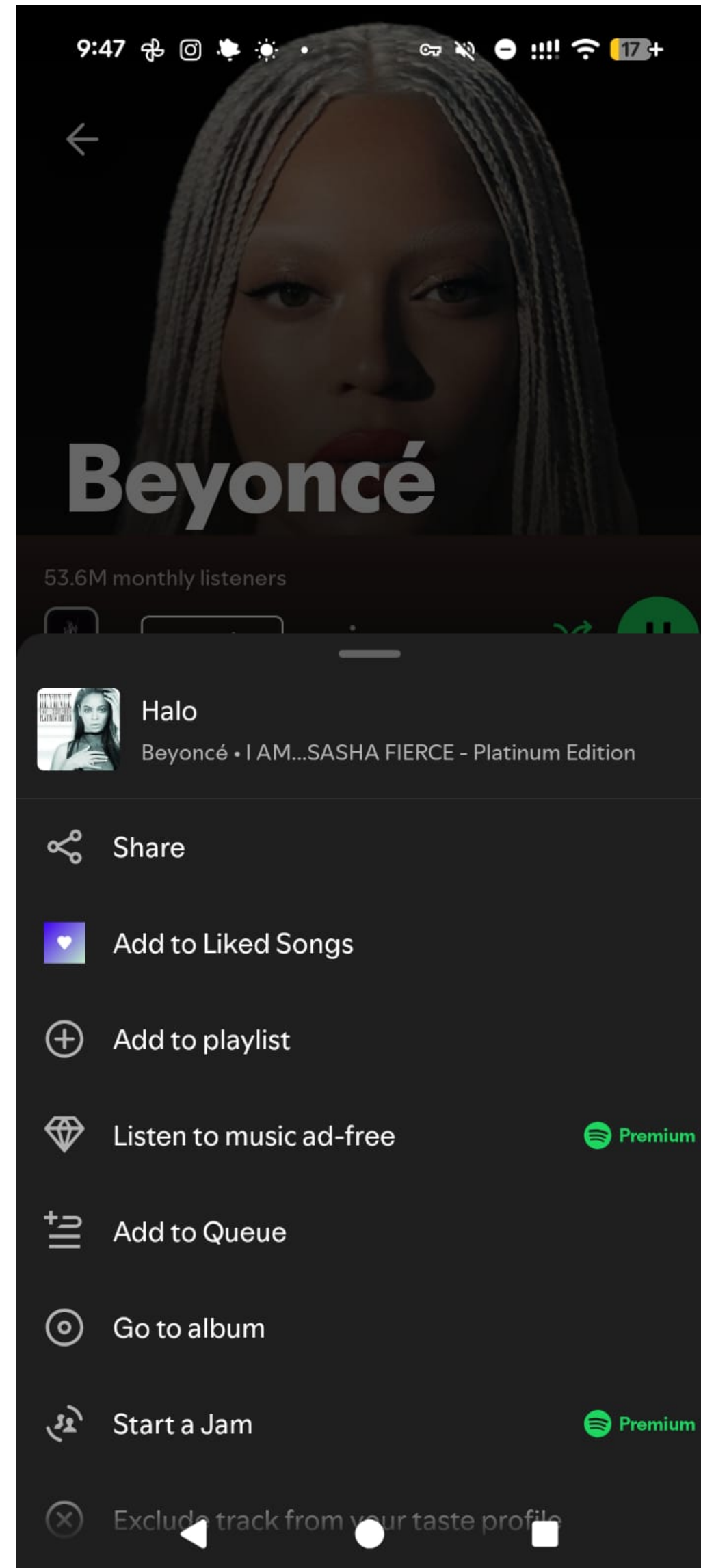


Grønbæk et al. MirrorBlender: Supporting Hybrid Meetings with a Malleable Video-Conferencing System. CHI '21. <https://doi.org/10.1145/3411764.3445698>

Polymorphism

Allowing the *same command* to affect *different types* of objects

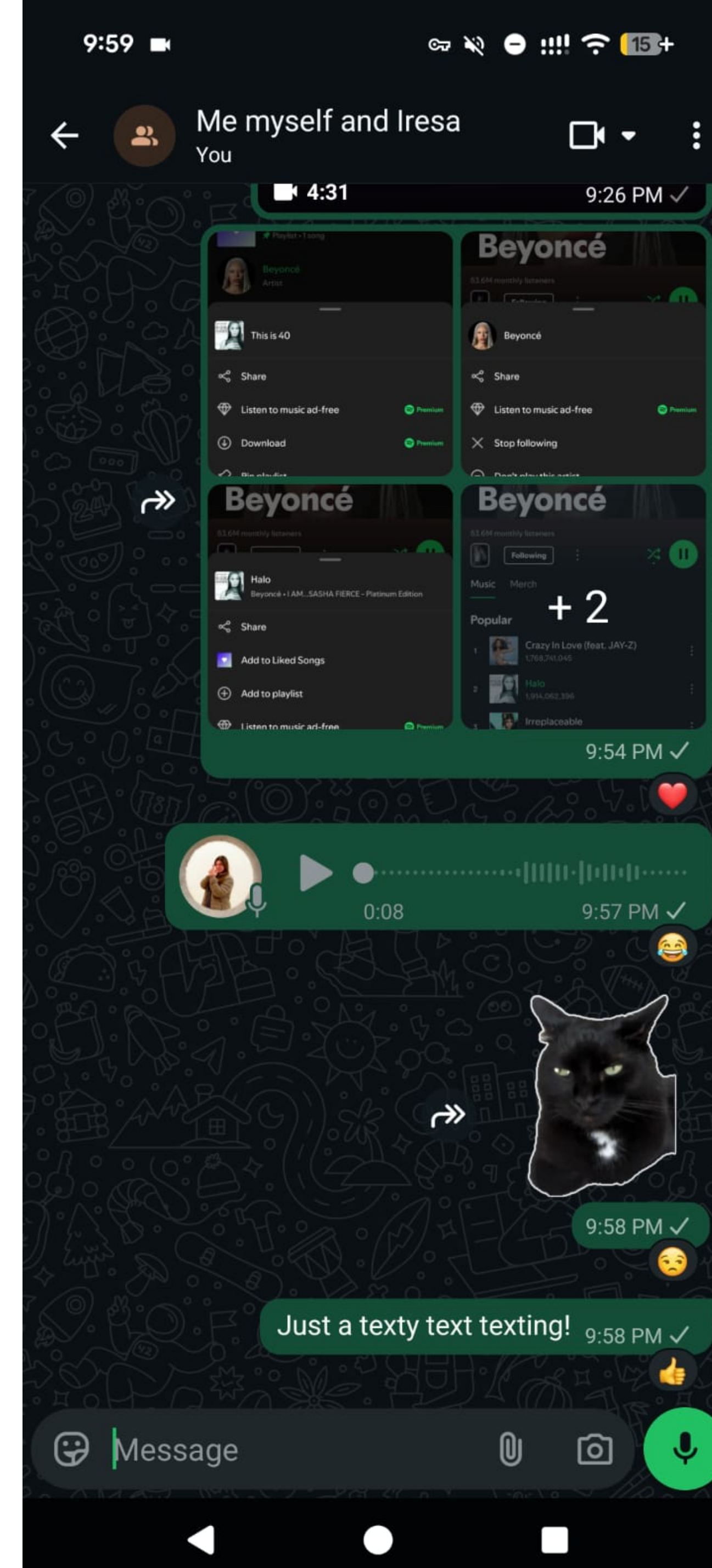
Example: The “share” command is *polymorphic* because it works with songs, artists and playlists



Polymorphism

Allowing the *same command* to affect *different types* of objects

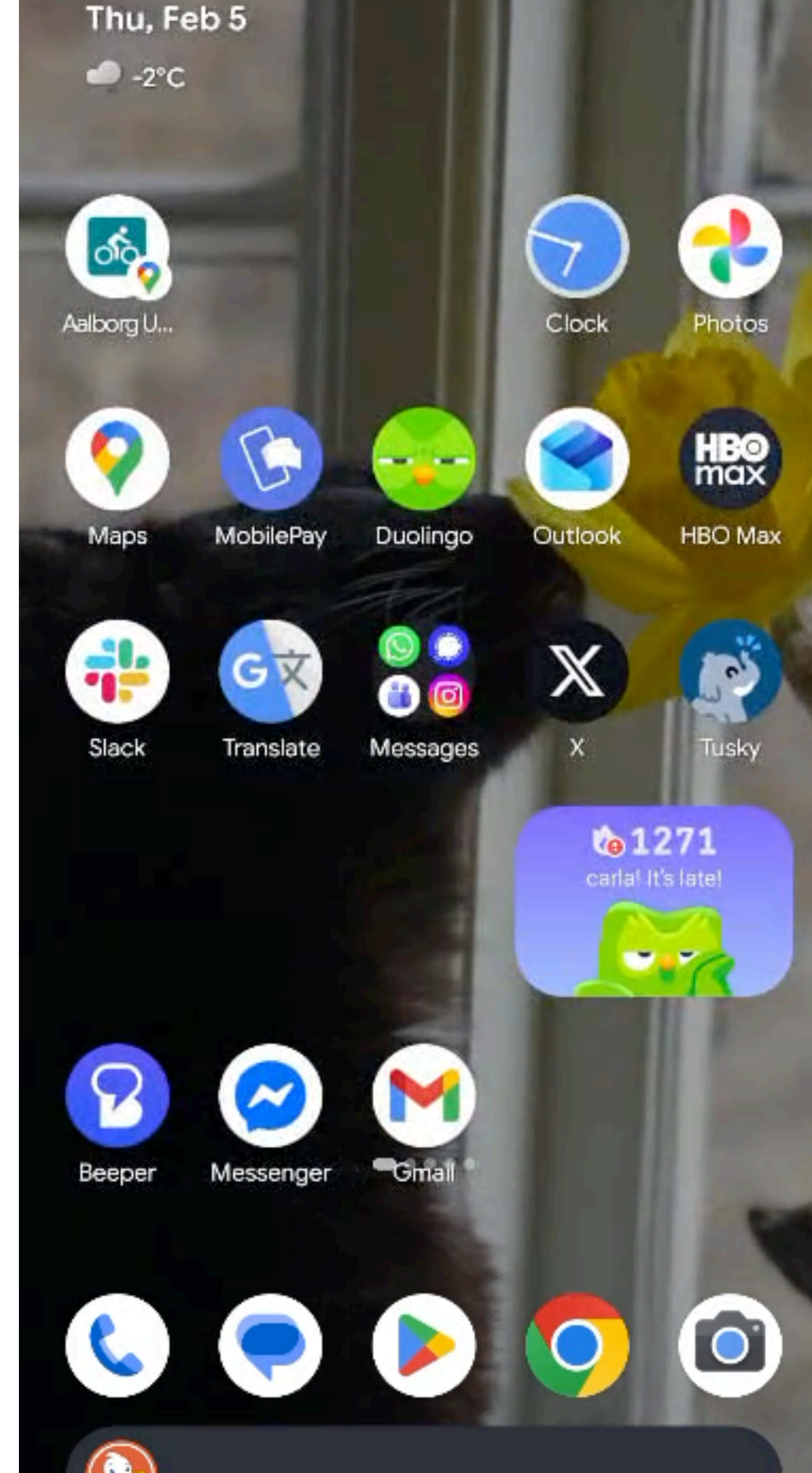
Example: *Reactions* on WhatsApp are *polymorphic* because I can use them on texts, stickers, pictures, groups of pictures and voice messages.



Polymorphism

Allowing the *same command* to affect *different types* of objects

Example: The “remove” command is *polymorphic* because it works with both apps and groups of apps

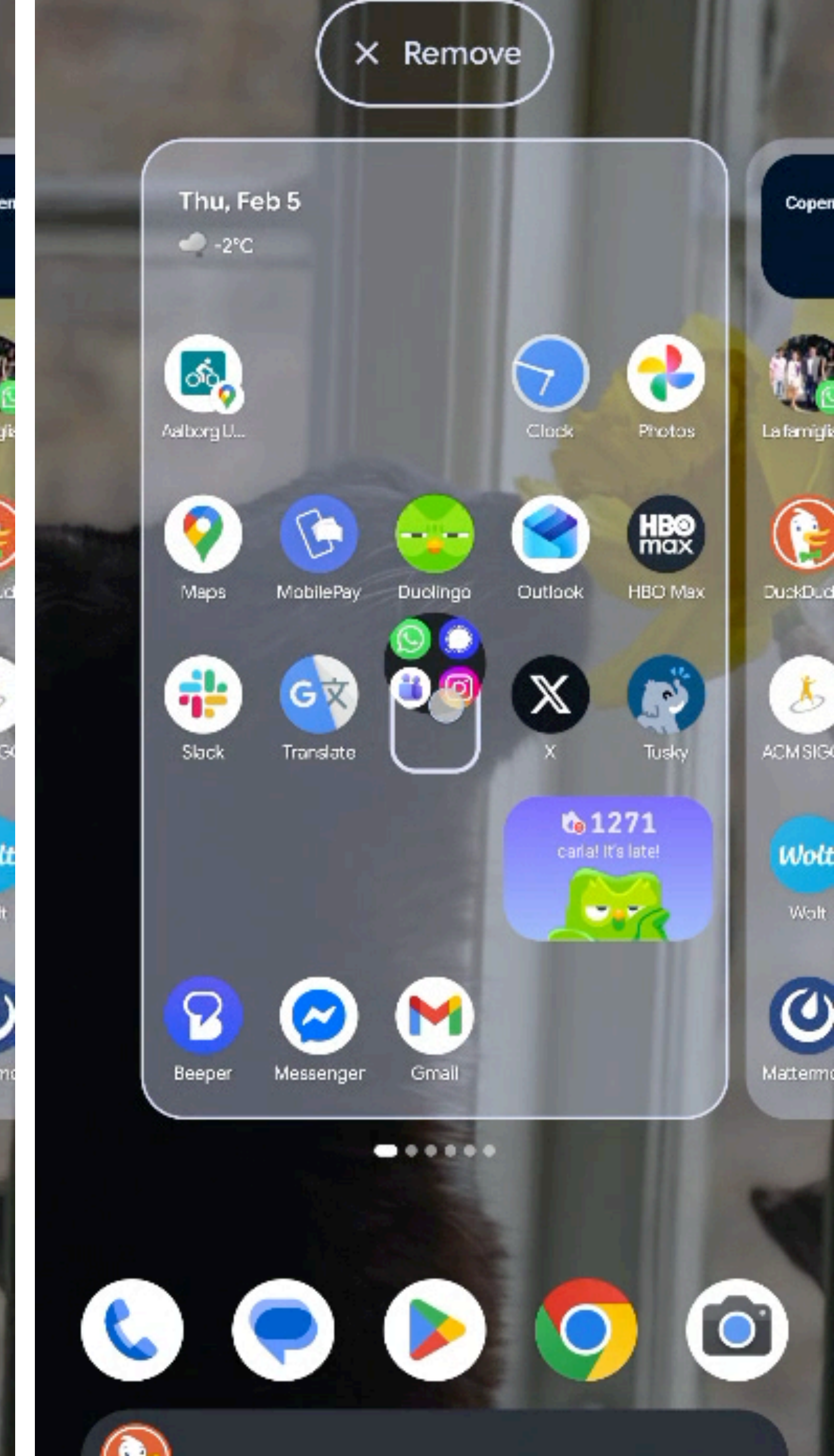
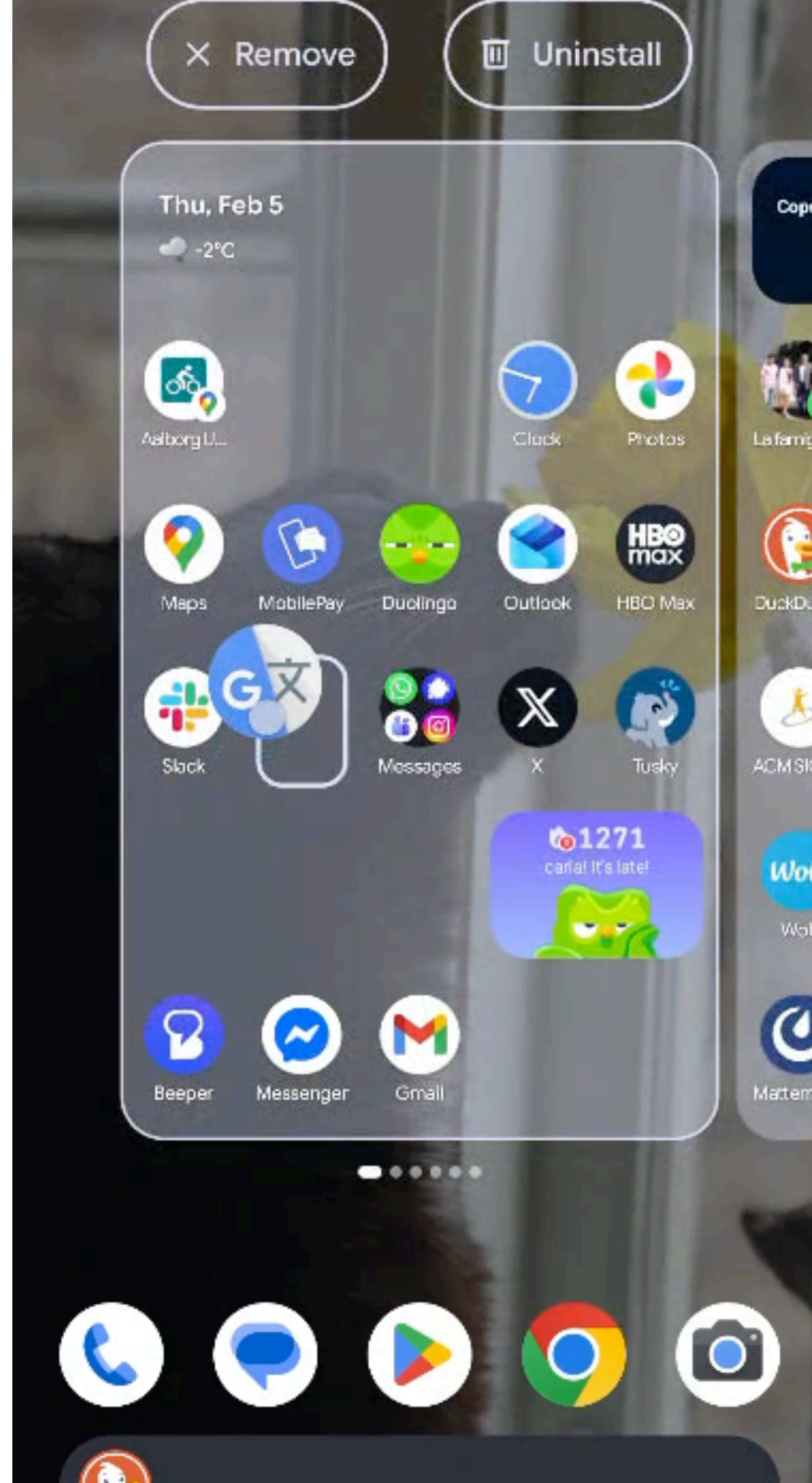


Polymorphism

Allowing the *same command* to affect *different types* of objects

Example: The “remove” command is *polymorphic* because it works with both apps and groups of apps

However, The “uninstall” command is *not* polymorphic, since it only works for apps

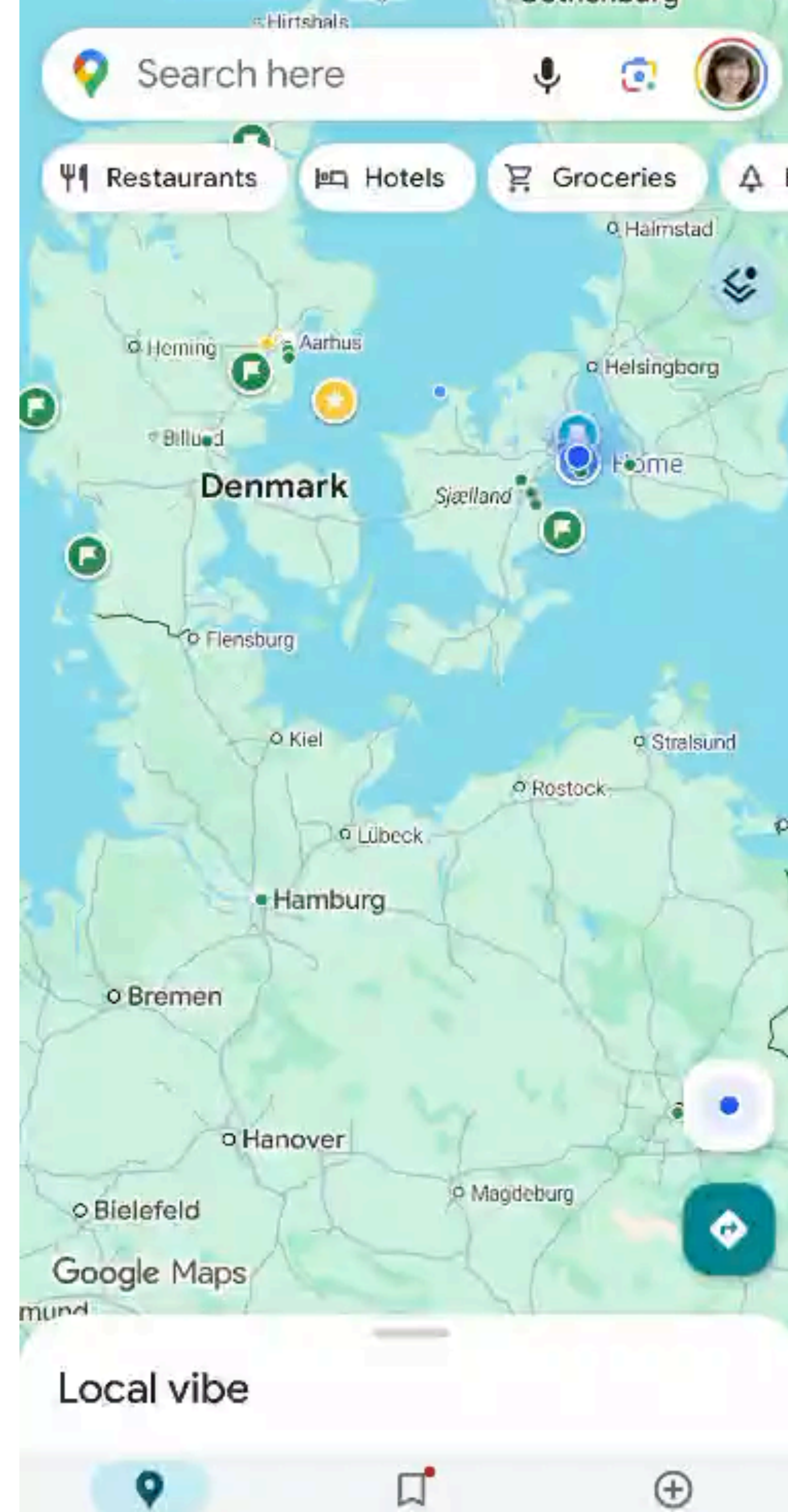


Reuse

Recycling *past* interactions
to optimize *future* actions

Recycling input

Example: The recent
search history allows users
to reuse their past input



Google Maps (Android)

Reuse

Recycling *past* interactions
to optimize *future* actions

Recycling input

Pini, Mikkel

3. User Control and Freedom An analysis of Ryanair

Good things

- **Independent form fields**, edit fields in any order
- **Swap Button** for quick reversal without retyping

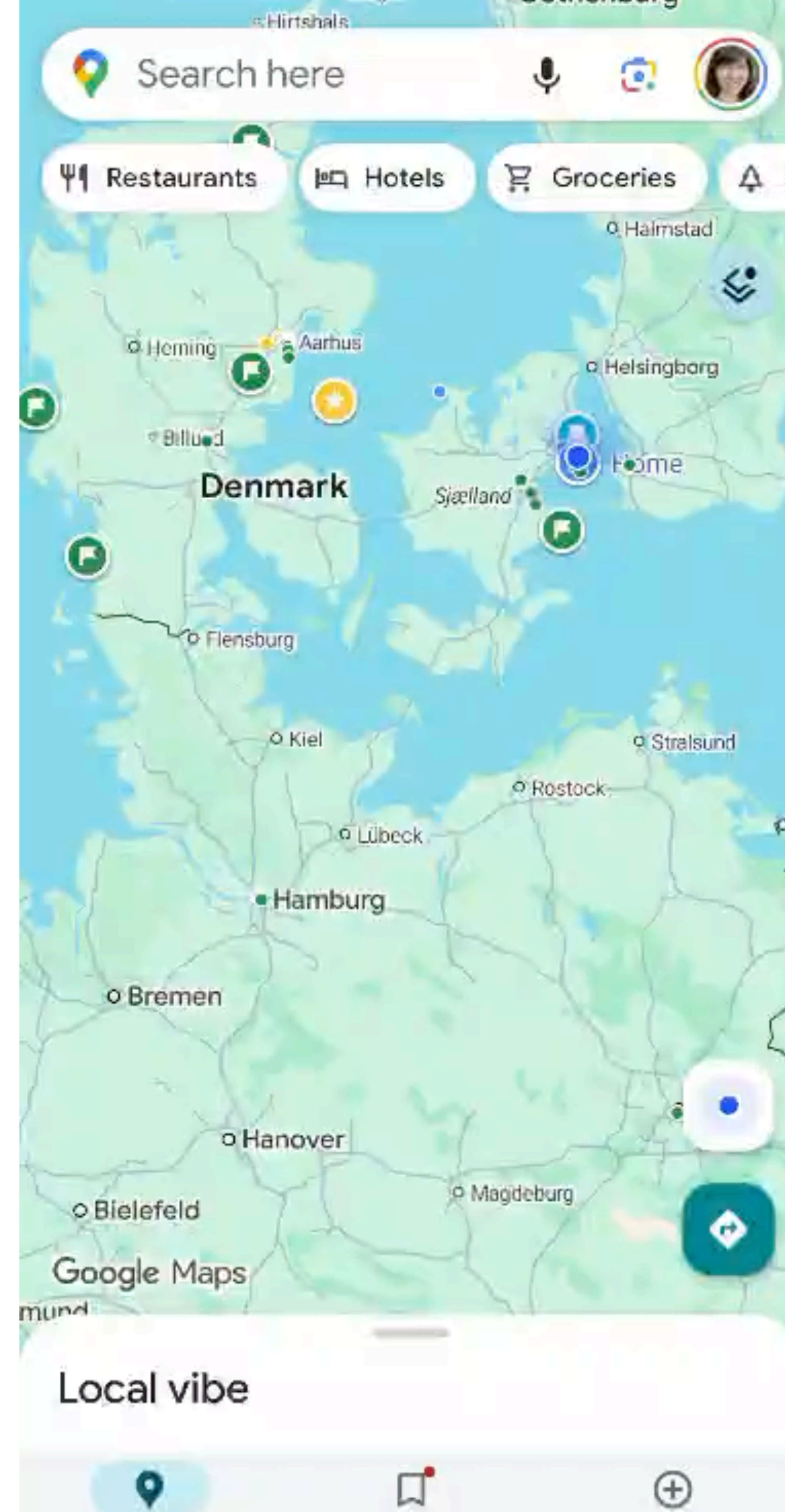
The image shows a screenshot of the Ryanair website's flight booking interface. A large pink arrow points down towards the interface. A circular callout highlights the 'Swap' button (two arrows forming a circle) located between the 'From' and 'To' fields. The interface shows a flight from Copenhagen to Malaga. Below the callout, the full booking form is visible, including trip type (Return trip), departure/return dates, and a calendar view for February 3rd to 5th. The flight details at the bottom show a 3h 45m flight from Copenhagen to Malaga.

Reuse

Recycling *past* interactions
to optimize *future* actions

Recycling input

Example: The “swap”
button allows users to
reuse their past origin and
destination *inputs* to see
the reverse route without
starting from scratch.



Google Maps (Android)

Reuse

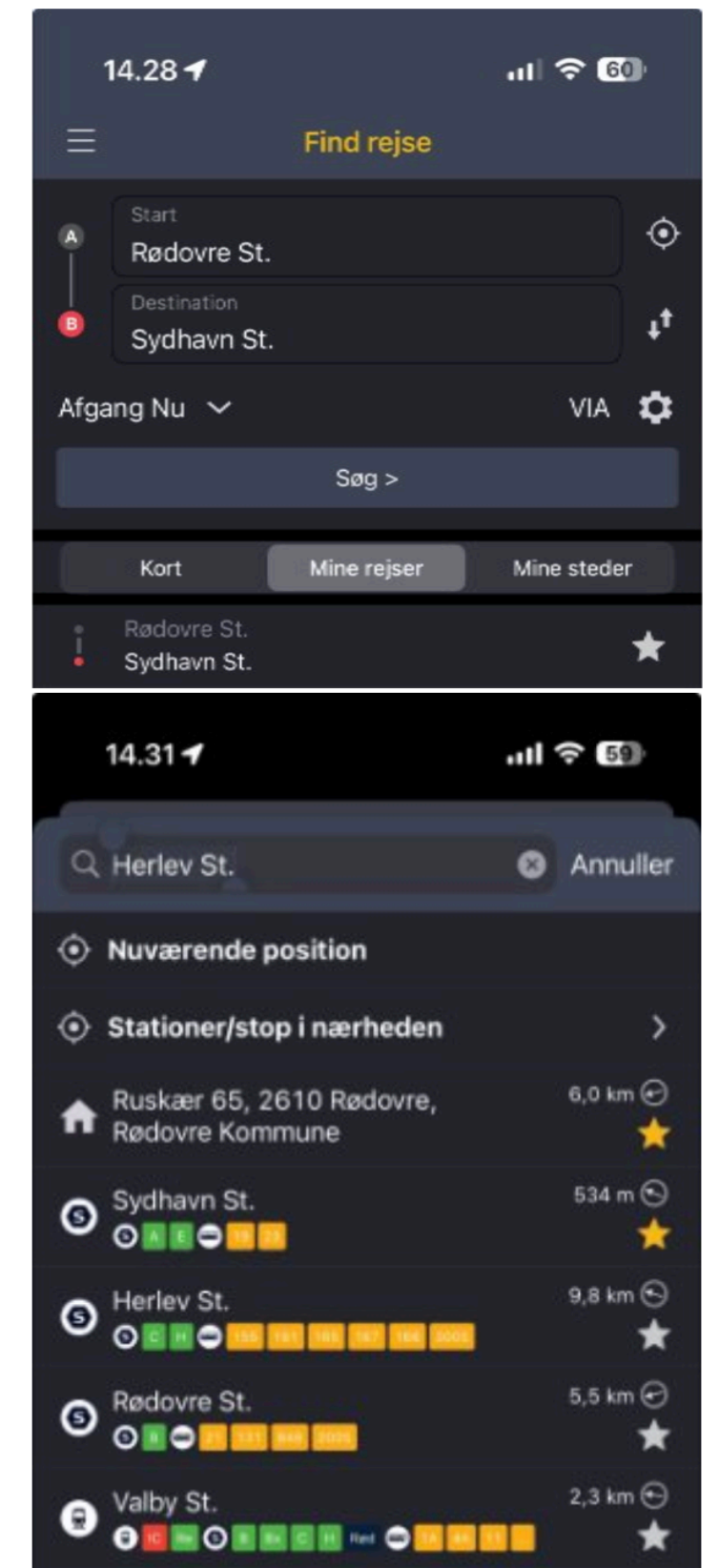
Recycling *past* interactions
to optimize *future* actions

Loke, Stina, Mathias

[Heuristic 7: Flexibility and Efficiency of use]

An analysis of Rejseplanen

- Holds
 - Remembers last tour and gives the ability to switch the destinations around after having been going from A to B.
 - Able to find your current location
 - When searching for stations, it lists last used stations as the top one
- Violates
 - Can only save destination and not route as favorite
 - Could suggest most used route based on personal use-case
 - Breaks multiple methods to accomplish the same task when you're not able to choose destination from a map.

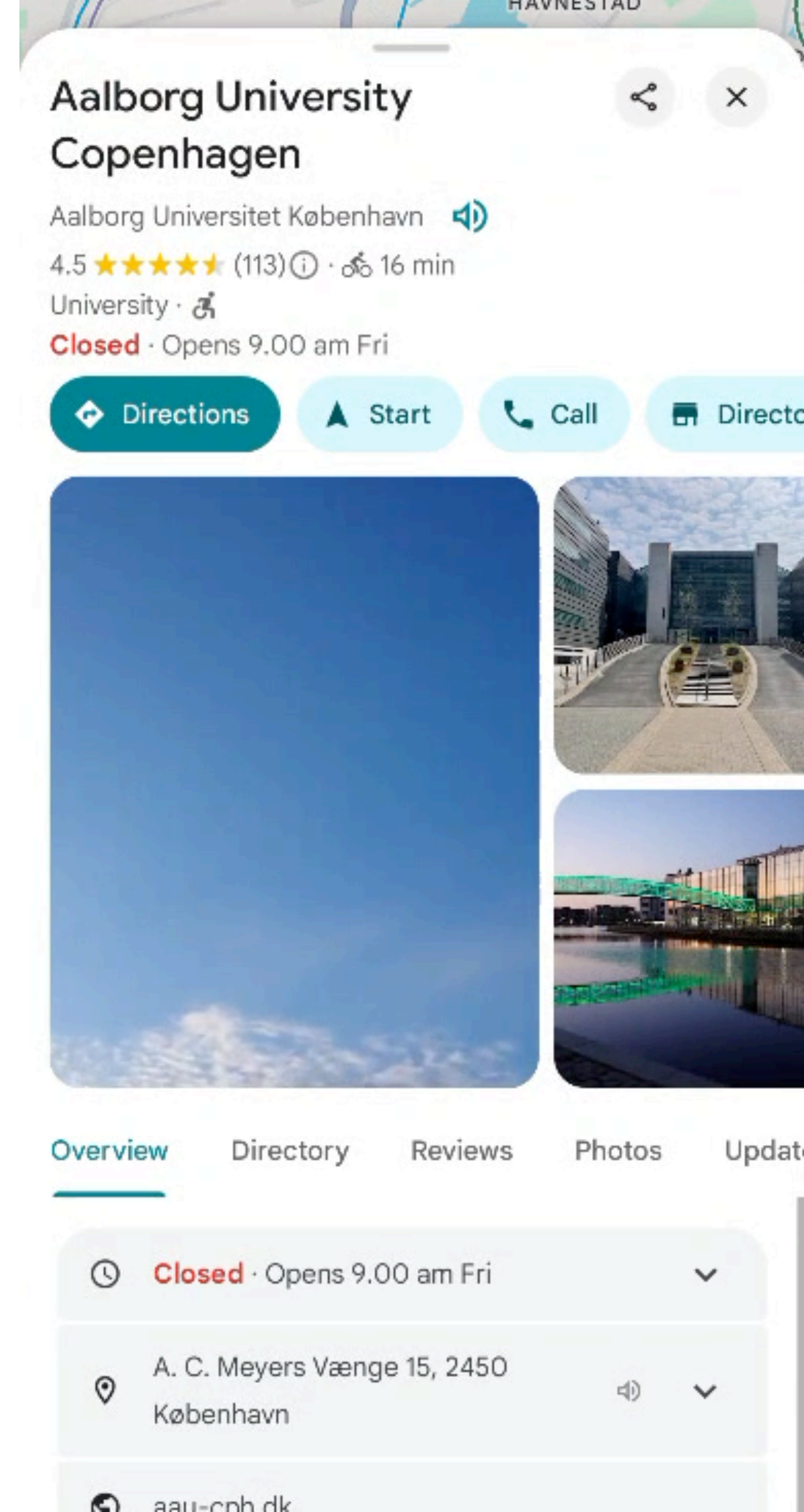


Reuse

Recycling *past* interactions
to optimize *future* actions

Recycling output

Example: *persisting* a
route as a shortcut to a
pre-defined route (e.g.,
Home -> Work)



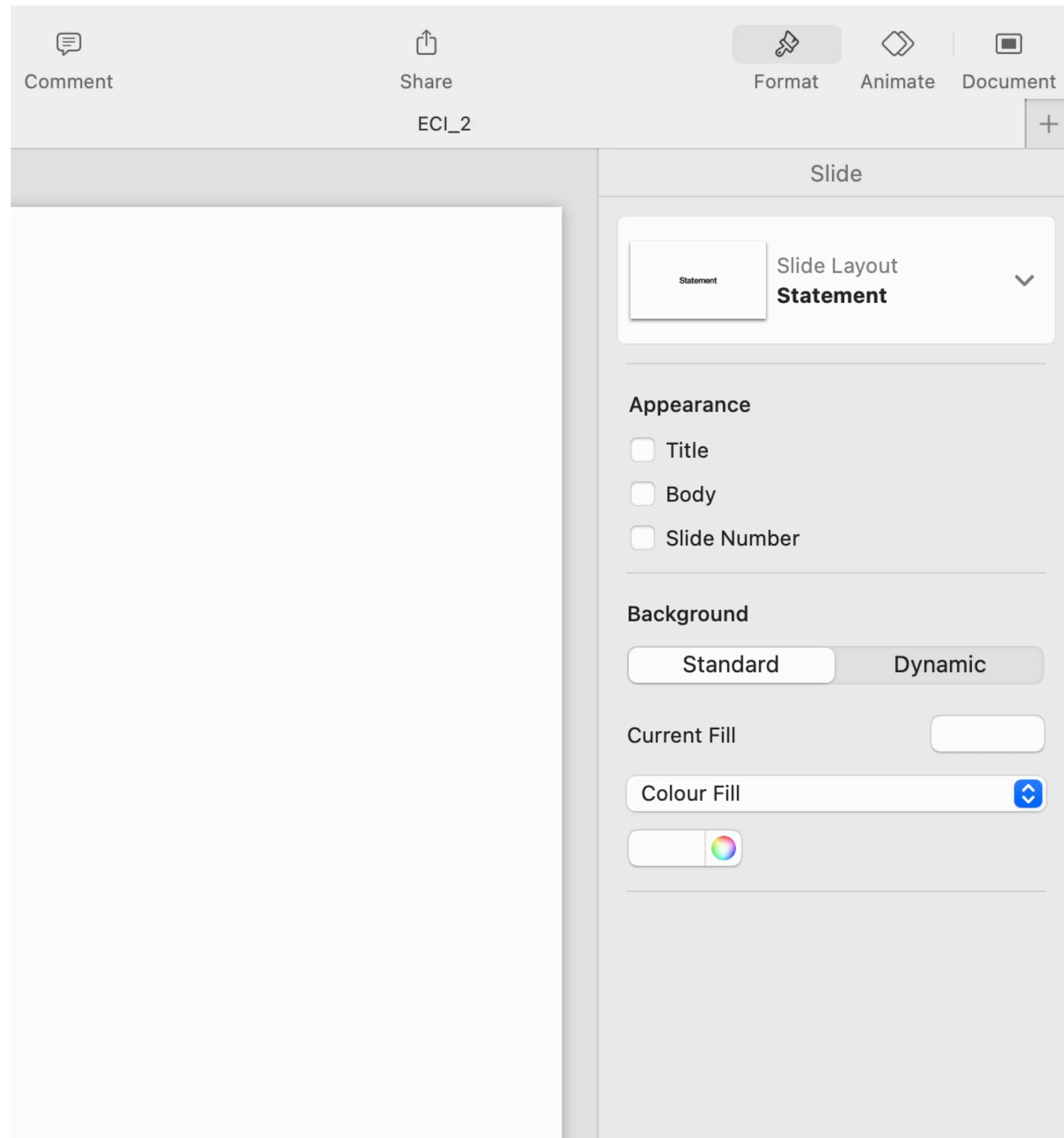
Reuse *(not mobile)*

Recycling *past* interactions
to optimize *future* actions

Recycling output

Example: *persisting* a
chosen color from a color
picker wheel as a
“swatch”

Color picker in MacOS



Reuse

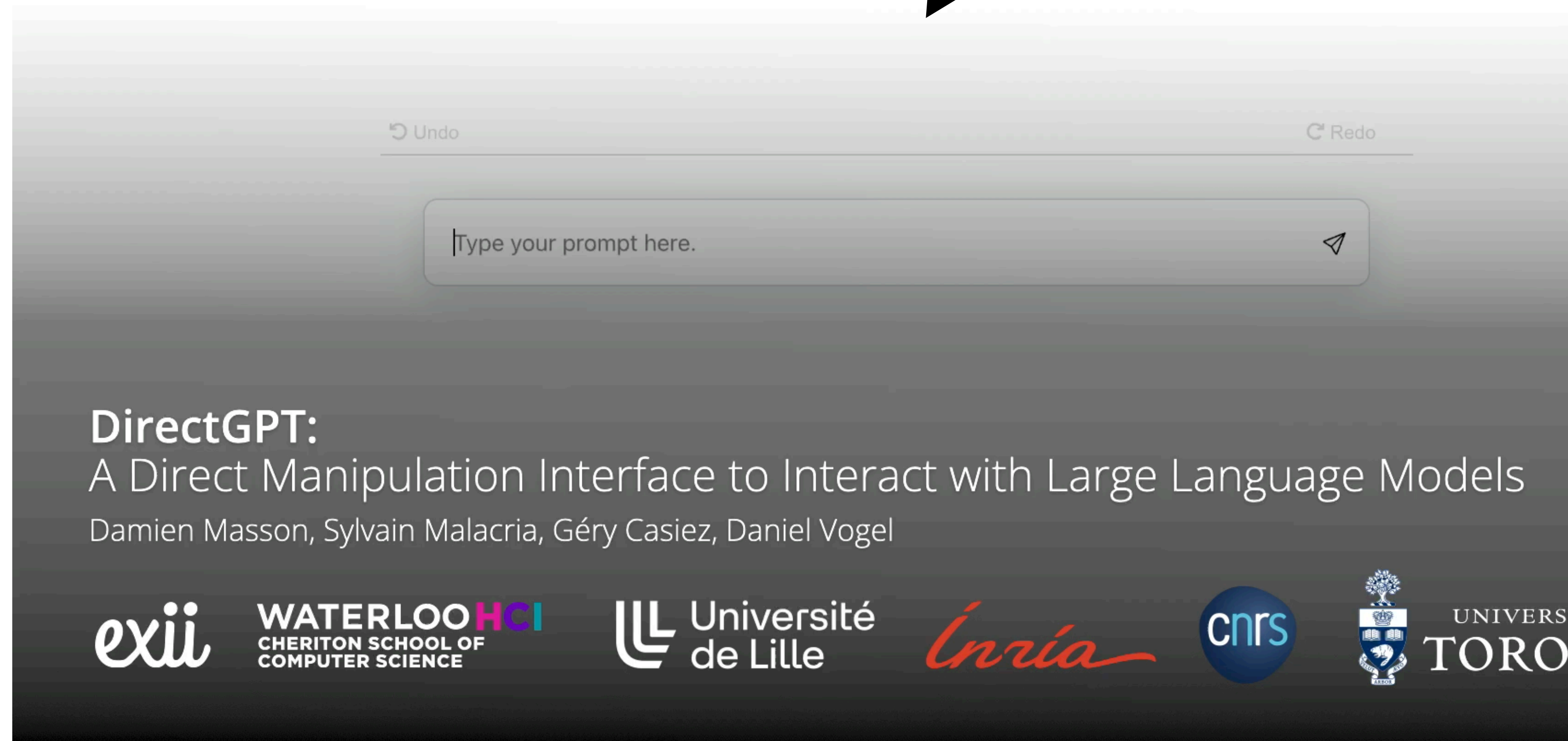
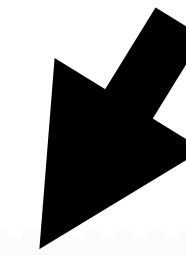
Other state-of-the-art examples outside of mobile devices

Recycling *past* interactions
to optimize *future* actions

Recycling output

Example:
reusing the output
of a ChatGPT-like prompt
as input
in the next prompt

Reified response (you can
interact with it) can be
used as input



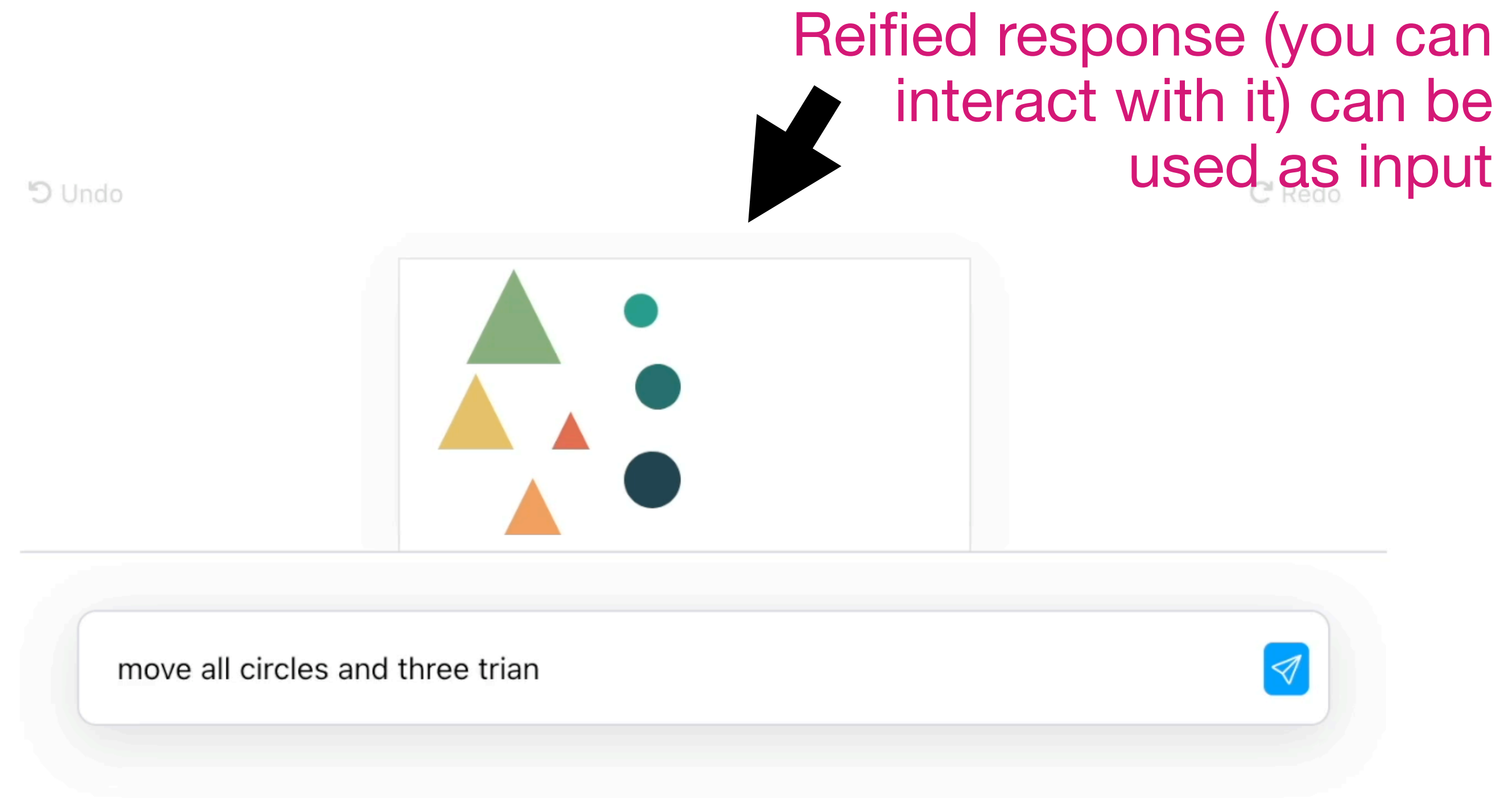
Reuse

Other state-of-the-art examples outside of mobile devices

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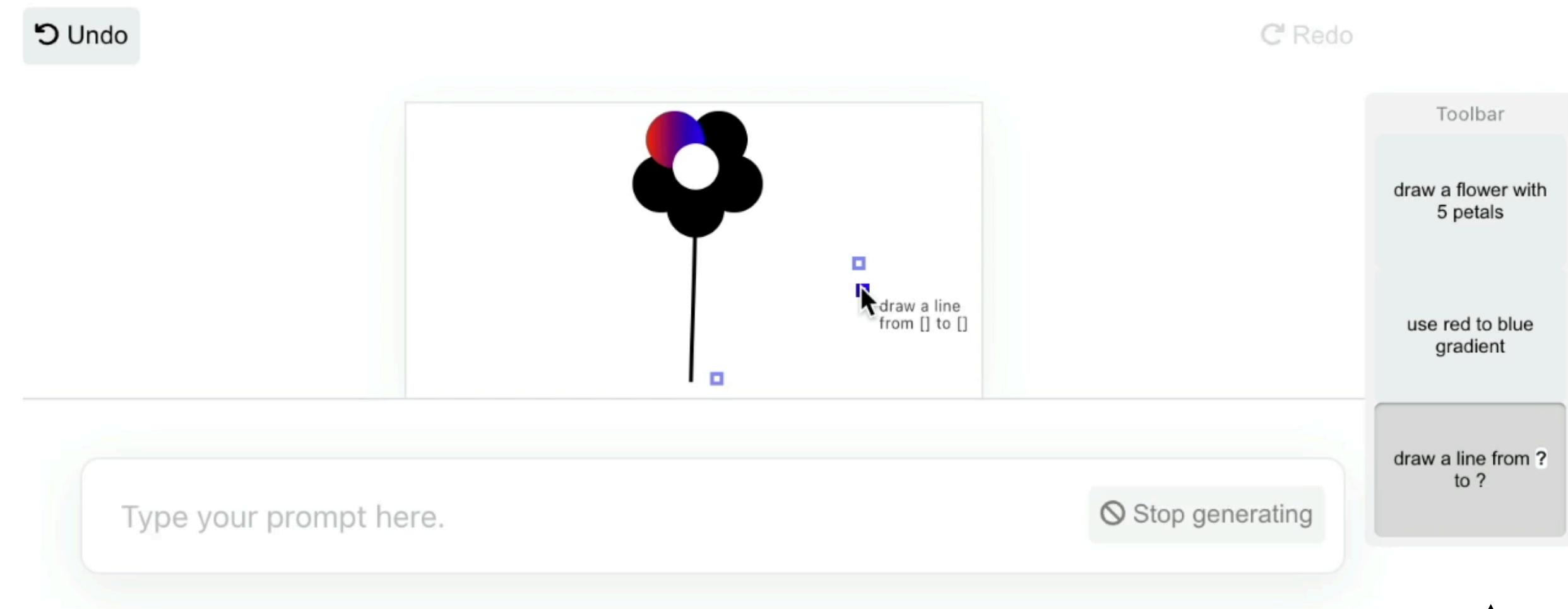
Reuse

Other state-of-the-art examples outside of mobile devices

Recycling *past* interactions
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Recycling output

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Reifying prompts (input) so
they can be reused!

Hands on!

Work in groups of 2-3

Present your answers here: https://aaudk-my.sharepoint.com/:p:/g/personal/wb25qs_cs_aau_dk/IQC_LyWJ-AYoQp9uMFrXOISoAQI4JWUfO7Hg5BqrsnnR3WY?e=Lgwpk4

1) Find 1 example of one or more principles (Reification, Polymorphism, Reuse) in your ***mobile*** apps or operating system. Explain it in the presentation.

(The same example can illustrate more than one principle, for example, a *reification* used for *reuse*)

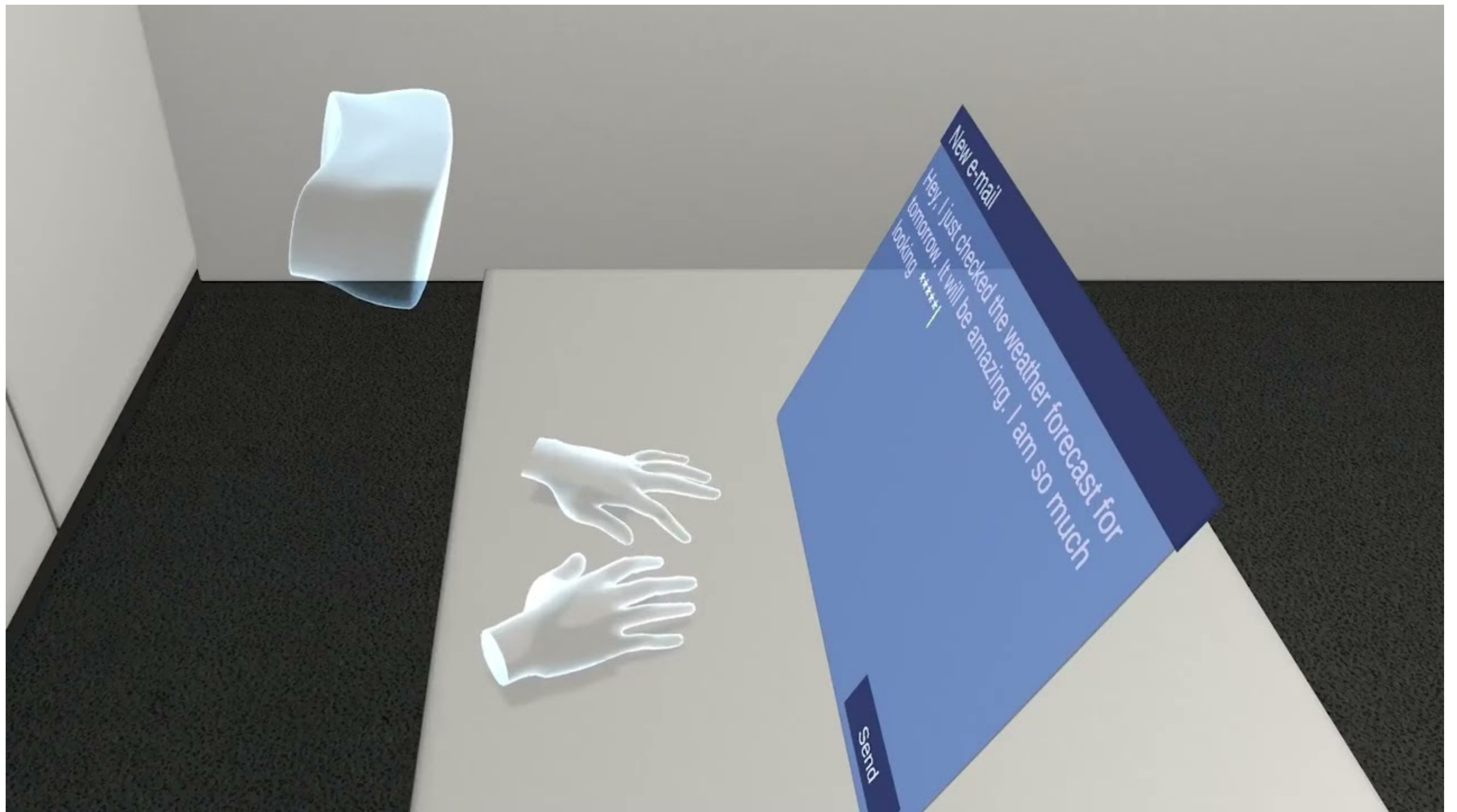
2) Find 1 example in your ***mobile*** apps or operating system of a design or interaction that DOESN'T apply any of these principles, but would benefit from them. Propose a new design that applies at least one principle.

Bonus Challenge: can you apply more than one principle to the same design?

Interaction Techniques

Interaction Techniques

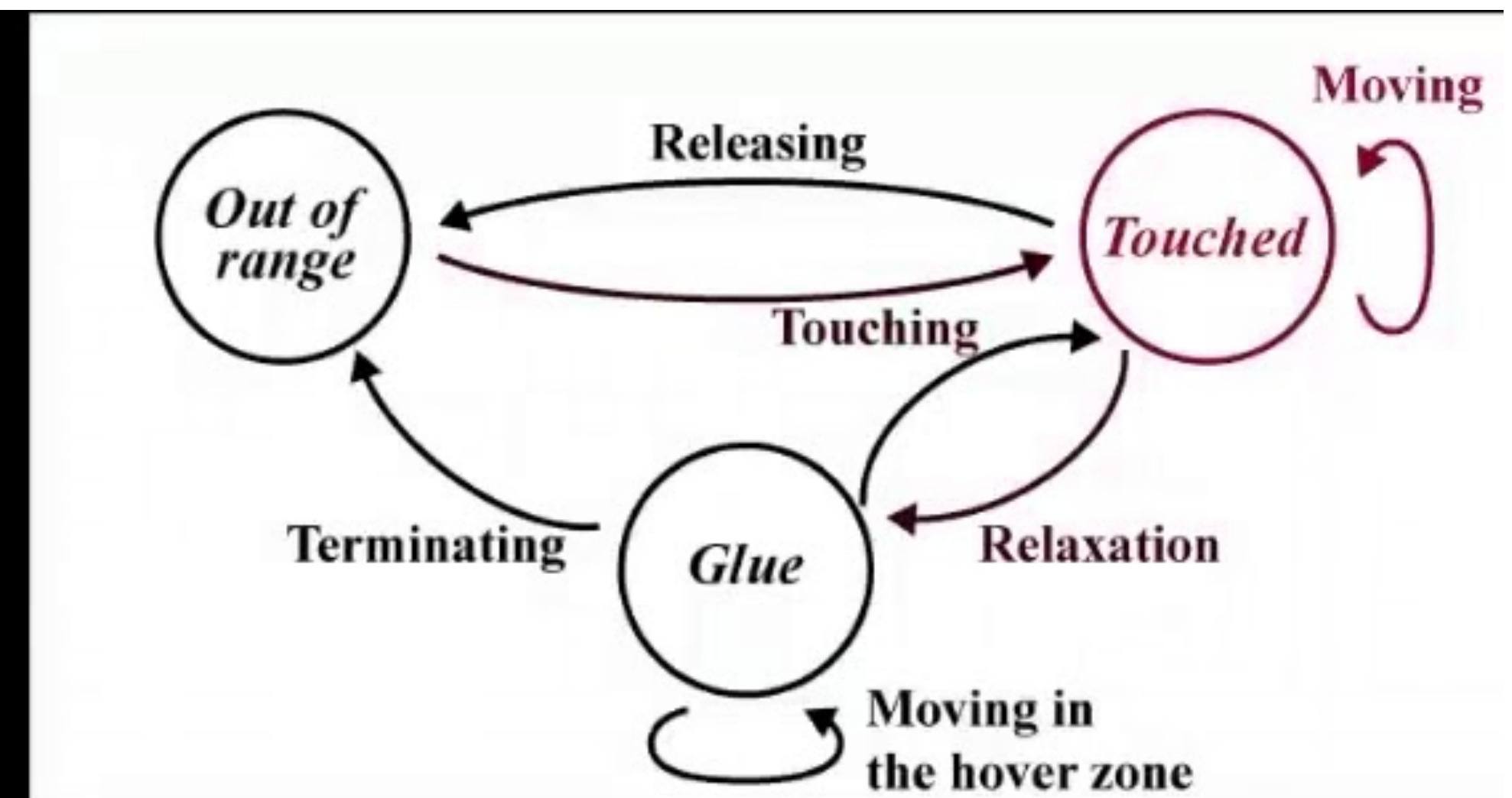
Technology-dependent methods for performing (often generic) interactions tasks
(e.g., typing, pointing, selecting)



TapType: <https://www.youtube.com/watch?v=iArXeN73UIU>



Relaxation action



Designing Touch Gestures Using the Space around the Smartwatch as Continuous Input Space:
<https://www.youtube.com/watch?v=fDc7Eru-SVE>

Interaction Qualities

What aspect (quality) of interaction do we want to improve with an interaction technique?

Examples:

Discoverability: How to help users discover available but hidden interactions and commands? (e.g., gestures)

Input Expressiveness: How to increase the range of available input possibilities?

Performance: How to help users perform tasks faster, or avoid errors?

and more!

Hands-on!

Choose an interaction technique to present and discuss here: https://aaudk-my.sharepoint.com/:p:/g/personal/wb25qs_cs_aau_dk/IQBdHidOp-BOT4wWouw4eurqAVI5oea2RO1BvpU8m73f_U8?e=tTKK4u